

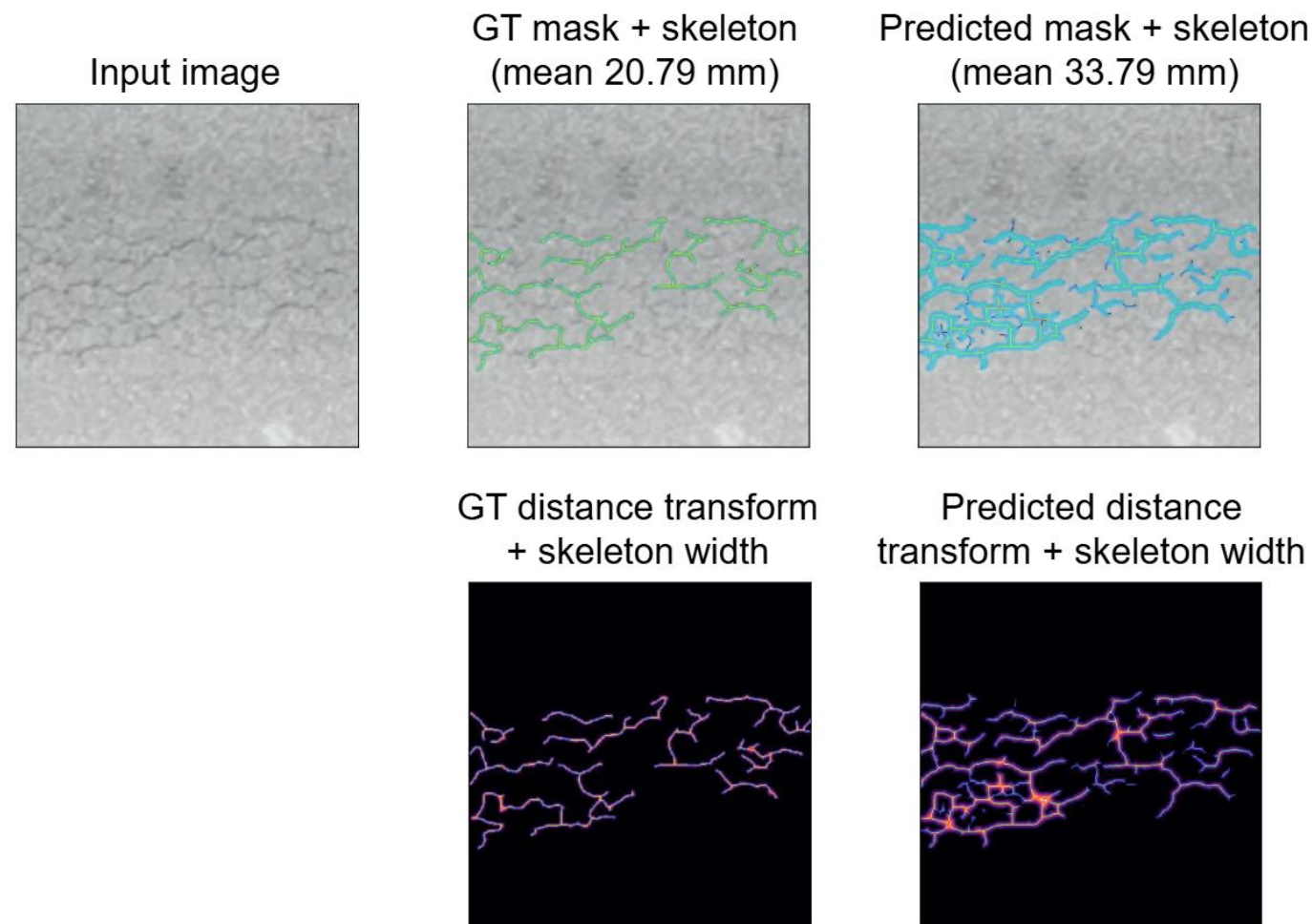


Figure 12: Discrepancies in width measurement arise due to differences in topological completeness. Although standard error metrics heavily penalize the model for overprediction, visual assessment indicates that the model identifies a more structurally realistic and interconnected fracture network, which is absent from the baseline manual labels. Note: Skeletonization reduces the segmented mask to a 1-pixel-wide centerline to represent the structural topology, and the distance transform calculates the local crack width by measuring the shortest Euclidean distance from this centerline to the nearest defect boundary.